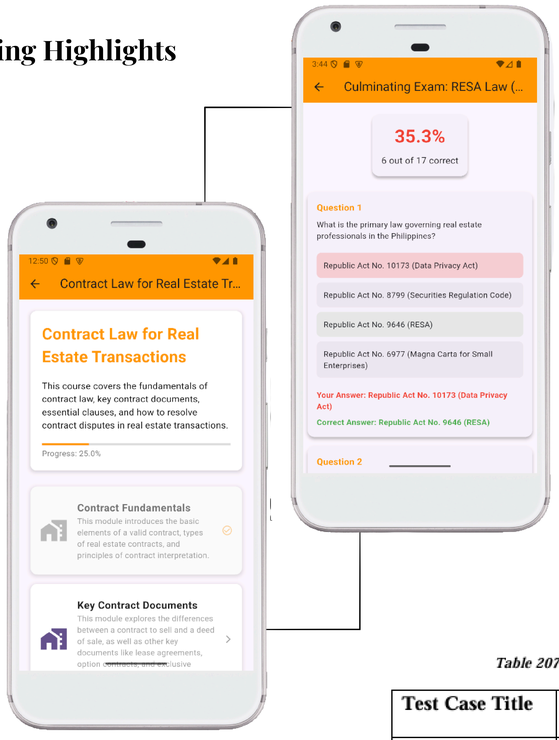
REdukasyon
Learning Management System

Tech Stack: Mobile Application (Flutter/Dart, PHP, MYSQL)

Overview: REdukasyon as a mobile platform aims to help real estate examinees prepare for licensure through structured courses, mock exams, and professional networking. I led the development of the Learning Management System (LMS), which powers the learning experience within the app. As the lead on the **learning management** side, I took a straightforward, test-as-you-build approach to ensure smooth integration across modules and functions.

Testing Approach: Manual testing methodology through an **end-to-end** perspective.

Implemented Features and End-to-End Testing Highlights
Development and Testing



Course Access and Learning Content

- Built a clean, scrollable course library showing users relevant real estate courses available.
- Designed smart search and filter so users can easily find courses by category, subscription type, or progress (e.g., In Progress, Completed).
- Implemented progress tracking through local caching and synced data across user sessions calculated as the ratio of completed modules to total modules per course.
- Allowed support for multiple content delivery: PDFs, slides, videos, and text, all viewable in-app without needing third-party apps.
- Ensured the app refreshed content when resumed and handled slow or failed internet connections gracefully.

End-To-End Testing Highlights:

- Tested full user flow from course viewing → course filtering → subscription check → content viewing → progress tracking.
- Simulated different user types (free vs. premium) and confirmed proper content access and upgrade prompts.
- Conducted multi-criteria filtering tests with combinations of date, subscription type, and progress status
- Tested progress synchronization by completing modules and verifying consistent progress display

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Test Case Title	Exan
Test Case No	TC-U
Preconditions	1. Us 2. Us 3. Us
Test Steps	1. Us 2. Th 3. Th Rang 4. Th 5. Th 6. Th for a
Expected Result	1. Th 2. Th Rang 3. W along
Actual Result	1. Th atten 2. Fil 3. Th ones
Status	Pass

Test cases can't be displayed fully due to **data privacy concerns**, but this is the prescribed **format**.

Examination and Performance Tracking

- Created timed and untimed mock exams with clear instructions and confirmation steps before starting.
- Added music and animations to make exams feel engaging and immersive.
- Showed users instant results, detailed answer breakdowns, and tips for improvement after each exam.
- Included a leaderboard so users could see how they ranked against others and stay motivated.
- Built a performance dashboard where users could track progress over time and identify weak areas.

End-To-End Testing Highlights:

- Tested full exam lifecycle: starting → answering → submitting → viewing results.
- Verified the correctness of score calculations, time tracking, as well as integration of question and choice randomization from the admin web side.
- Confirmed leaderboard updates and filters reflected user data accurately.
- Ensured performance charts, summaries and breakdown matched actual results.

PayMongo Integration (Payment API)

- Integrated PayMongo to process payments securely, using encrypted API connections.
- Generated unique payment links for each transaction, ensuring secure tracking through reference numbers.
- Built real-time checks and fallback updates so user subscriptions reflect payment status without delay.
- Embedded the payment page inside the app for a seamless experience without switching apps.

End-To-End Testing Highlights:

- Simulated full payment cycles from payment initiation → transaction → confirmation → subscription update
- Tested successful and failed payments and validated correct user subscription updates.
- Tested subscription plan handling with different durations (monthly, quarterly, annual) and validated automatic status updates for extensions, and expirations.

User Profile and Customization

- Built secure profile screens that load and display personal information, photos, and professional credentials.
- Allowed users to update their education, skills, occupation, and etc., easily through dynamic forms.

End-To-End Testing Highlights:

- Validated profile summary modal displays all user information correctly and completely
- Validated user flows for each section (e.g., education, occupation): Add → Edit → Delete → Save → View.
- Validated form validation logic by testing with valid, invalid, and edge case inputs